elmenue — Cost, Pricing and Revenue Report

Prepared 29 October 2026 · Written for: the owner, to study and approve every price before anything is bought.

A. The decision (read this first)

Choose this and settle on it:

Phase	Setup	Why
Now → about 15 busy cafés	**Vercel Pro (region fra1) + Supabase Pro (Frankfurt), with the adaptive-polling change (option B)**	No servers to run, nothing to patch, deployment is one command, and you already run on it. About $474/month at 50 cafés, $64 at 10. Frankfurt is the nearest Vercel region to Egypt that has regional pricing (others listed: Cape Town, Mumbai, Paris, London).
From about 15-20 busy cafés → final target	**Your own WebSocket server(s) on Hetzner (Germany) + Supabase Pro (option C), cafés split across servers by slug, Cloudflare in front**	Updates arrive instantly instead of every 2 s, the cost stops growing with polling (about $53/month at 50 cafés), and one server holds all 50 cafés' live state. It costs you operations time, so do it once you have revenue to justify a second machine and a checklist.

Why not option A or staying on polling forever: at 50 busy cafés it costs about $1,702/month, more than your subscription income. **Why not C on day one:** a solo owner running a restaurant platform on a single machine has no safety net yet; downtime during dinner service hurts every café at once. Managed hosting buys you that safety while the product is young. The offline queue already built (F7) keeps waiters ordering through short outages on either setup.

Hardware to standardise on (café buys it; you list it as a recommended kit): a Samsung Galaxy Tab A9+ as the till, an Xprinter 80 mm USB printer (Epson TM-T20III only for cafés that insist on a brand), an RJ11 cash drawer, a 650 VA UPS, a Lenovo Tab M10 as the kitchen screen, a TP-Link 4G backup router and two ceiling access points. **Buy one of each printer first and test it with the app before recommending anything:** the printing code uses WebUSB / WebSerial and I could not test a real printer.

Nothing has been bought or subscribed. I did not open any payment page or enter any card. Every figure below was read from the vendor's own page during this session (links in section 1) or is marked **ESTIMATE**. "Cost to you" is what leaves your bank account; "Client pays" is what the café pays someone, or what I recommend you charge the café.

How to read the numbers. Prices are in US dollars as the vendors quote them, with EGP shown at an **assumed EGP 50 per USD** (the sources I found disagreed on today's rate: one aggregator said about 31, another about 51; confirm on cbe.org.eg and rescale). All vendor prices exclude VAT unless stated; Egyptian VAT on your own service fees is a question for your accountant.

1. What I fetched (sources, all read on 29 Oct 2026)

Vendor / topic	Official page read	What it gave me
Vercel plans	vercel.com/pricing	Pro $20 per seat/month incl. $20 usage credit; 1 TB transfer; 10M CDN requests; $0.60/M invocations; add-ons
Vercel Frankfurt	vercel.com/docs/pricing/regional-pricing/fra1	fra1 rates: $2.60/M requests, $0.184/h CPU, $0.0152/GB-h memory, $0.15/GB transfer, $0.06/GB origin
Supabase	supabase.com/pricing	Pro $25; 250 GB egress then $0.09/GB; 8 GB DB then $0.125/GB; computes Micro $10 ... XL $210; PITR $100
Hetzner Cloud	hetzner.com/cloud/cost-optimized	CX23 €5.99, CX33 €8.99, CX43 €16.49 (incl. IPv4, 20 TB traffic). **Every plan was shown "currently unavailable" - check stock before planning on it**
Anthropic API	platform.claude.com/docs/en/about-claude/pricing	Sonnet 5.5 $2 in / $10 out per M tokens; Haiku 5.5 $0.10 / $0.50; Opus 5.5 $4 / $20; Fable 5.1 $10 / $50
Paymob Egypt	paymob.com/en/online-payment (FAQ)	2.75% + EGP 3 per successful card charge. No wallet rate, settlement time or refund fee is published: **ask Paymob for a written quote**
Egypt Trust via Orange	orange.eg/.../digital-seal-and-digital-signature	Digital seal (needed for e-invoicing) EGP 2,000 / 1 yr new, 1,800 renew; signature EGP 1,000 / 800; token re-issue 500; all incl. VAT
Cloudflare Registrar	cloudflare.com/products/registrar	Domains at registry cost, free WHOIS privacy; no .com price shown on the page (ESTIMATE about $10-12/yr)

2. What drives the cost: the load model

The system, as built for Vercel, has every tablet and phone ask the server for the current state every 2-2.5 seconds (Vercel cannot hold open connections). Each ask is billed three times: as a CDN request, a function run, and a database read. So cost follows **requests per month**, not tables or cafés directly.

Assumption (per café)	Value	Basis
Staff devices polling	5 every 2.2s	waiters, cashier, kitchen (code: 2 s captain/cashier, 2.5 s kitchen)
Guest phones at peak	45, half active	30 tables, about half full, 3 phones each
Peak requests/second	11.27	staff + guests
Average vs peak	35%	over a 12 h opening day (ESTIMATE)
Requests per café per month	5.1 million	3.9 req/s x 12 h x 30 days
Poll size	26 KB staff / 12.6 KB guest raw	**measured** on the demo café; about 5.0 KB after compression
Database row read per request	15 KB	**measured**: 12.9 KB venue data (sales history is stored separately)

These are estimates until a load test is run. If the real average is half of my assumption, every usage line in sections 3-4 halves; the fixed lines do not.

3.1 Monthly platform cost with 10 busy cafés (all 30 tables, Operations level)

Line	Cost to you / month (USD)	Notes
Option A - Vercel + Supabase as built		51M requests/month, 39 requests/s average
Vercel Pro seat	$20	1 seat; credit of $20 is subtracted below
CDN requests (10M incl., then $2.60/M in fra1)	$107	
Function invocations ($0.60/M)	$31	no Pro allowance listed on the pricing page
Fluid active CPU ($0.184/h)	$31	assumed 12 ms CPU per request
Provisioned memory ($0.0152/GB-h)	$17	assumed 1 GB x 80 ms per request
Fast data transfer + origin transfer	$15	256 GB; 1 TB included in transfer
Included usage credit	-$20	
Supabase Pro	$25	
Supabase compute (Large)	$100	$10 monthly credit already deducted
Supabase egress ($0.09/GB over 250 GB)	$47	767 GB read
Domain, other	$1	ESTIMATE
Total option A	**$374** (EGP 18,706)	per café: $37 / EGP 1,871
Option B - same stack, polling made adaptive (-70% requests)		15M requests/month (work to build: about one day)
Vercel (all lines, after $20 credit)	$33	requests $14, runs $9, CPU $5, memory $3, transfer $2
Supabase Pro + Small + egress	$30	egress 230 GB
Domain, other	$1	
Total option B	**$64** (EGP 3,182)	per café: $6 / EGP 318
Option C - own server (WebSocket mode) + Supabase		Phones keep one connection; the server pushes changes; database touched only on real changes
Hetzner CX33 (4 vCPU / 8 GB) x 1	$10	€8.99 each at assumed $1.10/€; **stock to be checked**
Server backups	$1	from €1.20 each (page: "starting at")
Supabase Pro + small compute	$25	low traffic: writes only
Domain, Cloudflare free plan	$1	ESTIMATE
Total option C	**$37** (EGP 1,860)	per café: $4 / EGP 186. Plus your time for updates and uptime

Option C trade-off: you operate it. Keep a second server and nightly backups; one process serves all cafés, so test restarts before opening it to paying venues.

3.2 Monthly platform cost with 50 busy cafés (all 30 tables, Operations level)

Line	Cost to you / month (USD)	Notes
Option A - Vercel + Supabase as built		256M requests/month, 197 requests/s average
Vercel Pro seat	$20	1 seat; credit of $20 is subtracted below
CDN requests (10M incl., then $2.60/M in fra1)	$639	
Function invocations ($0.60/M)	$153	no Pro allowance listed on the pricing page
Fluid active CPU ($0.184/h)	$157	assumed 12 ms CPU per request
Provisioned memory ($0.0152/GB-h)	$86	assumed 1 GB x 80 ms per request
Fast data transfer + origin transfer	$118	1,278 GB; 1 TB included in transfer
Included usage credit	-$20	
Supabase Pro	$25	
Supabase compute (XL)	$200	$10 monthly credit already deducted
Supabase egress ($0.09/GB over 250 GB)	$323	3,835 GB read
Domain, other	$1	ESTIMATE
Total option A	**$1,702** (EGP 85,120)	per café: $34 / EGP 1,702
Option B - same stack, polling made adaptive (-70% requests)		77M requests/month (work to build: about one day)
Vercel (all lines, after $20 credit)	$267	requests $173, runs $46, CPU $24, memory $16, transfer $8
Supabase Pro + Large + egress	$206	egress 1,150 GB
Domain, other	$1	
Total option B	**$474** (EGP 23,713)	per café: $9 / EGP 474
Option C - own server (WebSocket mode) + Supabase		Phones keep one connection; the server pushes changes; database touched only on real changes
Hetzner CX33 (4 vCPU / 8 GB) x 2	$20	€8.99 each at assumed $1.10/€; **stock to be checked**
Server backups	$3	from €1.20 each (page: "starting at")
Supabase Pro + small compute	$30	low traffic: writes only
Domain, Cloudflare free plan	$1	ESTIMATE
Total option C	**$53** (EGP 2,671)	per café: $1 / EGP 53. Plus your time for updates and uptime

Option C trade-off: you operate it. Keep a second server and nightly backups; one process serves all cafés, so test restarts before opening it to paying venues.

4. How to price the product so it cannot lose money

Formula. Price per café per month = (hosting cost per café + AI cost + support allowance + payment-processing cost you carry) / (1 - target margin). Your own time and tax come after that.

Hosting option	Hosting cost per busy café	Price needed at 50% margin	Price needed at 70% margin
A: as built	$34 = EGP 1,702	EGP 3,405	EGP 5,675
B: adaptive polling	$9 = EGP 474	EGP 949	EGP 1,581
C: own server	$1 = EGP 53	EGP 107	EGP 178

With your current list prices (Basic EGP 350-500, Operations EGP 850-1,000, VIP EGP 1,450) **option A loses money on every busy café** (cost EGP 1,702 per café). Option B is profitable. Option C leaves almost all of the price as margin.

4.1 Example: 50 paying cafés (35 Operations at EGP 950, 15 VIP at EGP 1,450)

	Option A	Option B	Option C
Subscription revenue / month	EGP 55,000 ($1,100)		
Platform cost / month	$1,702	$474	$53
AI cost, 15 VIP cafés (Haiku 5.5)	$3	$3	$3
Gross profit / month (USD)	$-606	$623	$1,043
Gross profit / month (EGP)	EGP -30,277	EGP 31,130	EGP 52,172
Gross margin	-55%	57%	95%

Before this is "your income": subtract income tax and VAT treatment (accountant), your time for support, bank and transfer fees, and any money you pay people. Revenue also depends on how many cafés really pay on time: model 10% of cafés late or churned.

4.2 Break-even

Fixed floor under option C is about $53/month, i.e. **one or two Operations cafés** pay for the hosting. Under option A the cost grows with every café (about $34 each), so it never gets cheaper per café; under B about $9 each; under C it falls toward $1-2 per café.

4.3 Basic-tier cafés are almost free to host

A Basic café (QR menu only) has no polling tablets and guests fetch the menu once per scan: roughly 10,000 requests a month, well under $1 of hosting. They are a pure-margin tier; the price of EGP 350-500 mostly pays for support.

5. Extras and add-ons a café may ask for: what it costs you, what the café pays

Add-on	Cost to YOU	What the CAFÉ pays (vendor)	Suggested price to charge the café
Online card / wallet payments (Paymob)	No per-payment cost; a few cents of hosting per thousand payments (ESTIMATE)	**Paymob: 2.75% + EGP 3 per card charge** (published); wallet rate not published, ask Paymob. Example: EGP 200 bill = EGP 8.50 fee	Optional: free, or EGP 150/month, or 0.3-0.5% of online volume as a platform fee. Do not add it silently to the guest's bill
ETA e-receipts	Hosting is negligible. Your cost is setup and support time	**Digital seal (Egypt Trust via Orange): EGP 2,000 / 1 yr new, EGP 1,800 renew, incl. VAT** (about EGP 167/month). ETA itself: no API fee found	Setup EGP 1,000-2,500 once; EGP 150-250/month for monitoring and retries
AI assistant (VIP) - Sonnet 5.5 as coded	$0.014 per question (EGP 0.70); 300 questions = $4.20 / month	-	Included in VIP
AI assistant - Haiku 5.5 (set env ANTHROPIC_MODEL)	$0.0007 per question; 300 questions = $0.21 / month	-	Same: included; about 20x cheaper than Sonnet
AI menu photo scan	$0.023 per scan (Sonnet) / $0.0011 (Haiku)	-	First 3 scans free, then EGP 20 per scan or part of onboarding
Extra branch	Under $1/month of hosting	-	EGP 400-600 per branch per month (VIP includes 3)
More tables / staff than the plan limit	Under $1/month	-	Upgrade to the next tier, or EGP 15-25 per extra table per month
Custom domain for the café	Domain registration, about $10-12 / year (ESTIMATE); Vercel adds no fee for the domain	Registrar price	EGP 600-900 per year including setup
Menu entry / photo digitising	Your time (plus AI scan cost above)	-	EGP 500-2,000 one-time depending on menu size
On-site onboarding and staff training	Your time and transport	-	EGP 1,000-3,000 one-time
Printed table QR stands	**No price found**: get a local print-shop quote	Print cost	Cost + 30-50% pass-through
80 mm thermal printer + cash drawer	**No price found**: get a local quote	Hardware	Pass-through at cost + 10-15%, or let the café buy it
Extended database recovery (Supabase PITR)	$100/month per 7 days of history, for the whole platform, not per café	-	Only add once you have paying cafés that ask; spread over all cafés
Priority support	Your time	-	EGP 300-500/month if you want to sell it (WhatsApp channel is out of scope in this release)

Prices in the last column are **recommendations for you to decide**, based on typical practice; I have no market data for Egyptian competitors beyond what is in your master plan. Confirm with 3-5 cafés before publishing a price list.

6. What you pay, when: the budget to approve

Item	When	Option A (50 cafés)	Option B (50 cafés)	Option C (50 cafés)
Platform hosting	monthly	$1,702	$474	$53
Domain	yearly	$12	$12	$12
AI usage (15 VIP, Haiku)	monthly	$3	$3	$3
Monthly total		**$1,707**	**$478**	**$58**
First-year cash (12 months + domain)		**$20,479**	**$5,741**	**$691**

Pilot (first 10 cafés) on option A: $374/month; on option B: $64/month; on option C: $37/month. The Vercel and Supabase free plans cannot be used for paying cafés (Vercel Hobby is non-commercial per its own FAQ; Supabase Free is 500 MB).

Paid by each café, not by you: Paymob fees (2.75% + EGP 3), the e-signature / digital seal (EGP 2,000 first year), printers, QR printing.

7. Authorisation checklist (nothing is purchased until you tick and tell me)

	Item	Cost to you	Notes
■	Vercel Pro (1 seat), region fra1	$20/month + usage	needed for options A and B
■	Supabase Pro, Frankfurt, one project	$25/month + compute	all options
■	Hetzner server(s) CX33	€8.99 each/month	option C only; verify stock first
■	Domain (Cloudflare Registrar)	about $10-12/year	ESTIMATE until you see the price at checkout
■	Anthropic API credit (pay as you go)	start with $20	cost grows only with AI use
■	Paymob merchant account (per café)	no monthly fee published	get the written rate for wallets and settlement time
■	ETA digital seal (per café)	EGP 2,000 first year	paid by the café

8. Things I could not confirm (do these before you commit)

1) Today's EGP/USD rate. 2) Hetzner stock (all plans showed unavailable). 3) Paymob wallet fee, settlement time and refund fee: only the card rate is published. 4) Whether the Vercel included $20 credit and the 10M-request allowance apply exactly as the pages state when billed in fra1. 5) Real load: run a one-week simulated load test to replace my assumptions in section 2. 6) Egyptian VAT and income-tax treatment of your fees. 7) Printing and hardware prices (no source found).

My recommendation: ship the polling reduction (option B) before the first paying café, run a load test, and plan the move to option C once you pass 15-20 busy cafés. Price Operations at EGP 950+ and keep VIP at EGP 1,450+ unless your accountant says otherwise.

9. Hardware: prices found, per item (cost to YOU = EGP 0 unless you resell)

Source: Egyptian retailers read on 29 Oct 2026 through web search (Amazon.eg, Noon, Jumia, 2B, Egypt Laptop, CairoPOS and others; links in the appendix). I could not open the product pages in a logged-in shopping session, so each number is a **listing snapshot**; samsung.com/eg returned an empty tablet list, so Samsung prices are retailer prices. Prices include Egyptian VAT as retailers show them. The café pays the retailer; you pay nothing unless you decide to resell.

Item	Model	Price range found	Where found	Confidence
Cashier tablet 11"	Samsung Galaxy Tab A9+ (4-8 GB / 64-128 GB)	EGP 11,299 - 13,918	Amazon.eg 13,918; Noon 12,499-13,153; Ennap 11,299	Medium: retailer listings; samsung.com/eg showed no prices; one tracker said 8,810 (not used)
Tablet 8.7" (kitchen / spare)	Samsung Galaxy Tab A9 (4 GB / 64 GB)	EGP 7,100 - 8,479	Ehab Group 8,479; Ennap 7,299; tv-it 7,100	Medium
Budget cashier / kitchen tablet	Lenovo Tab M10 3rd gen 4G (4 GB / 64 GB)	EGP 6,299 - 6,999	XPRS 6,299; Noon M10 X505X 6,999	Medium
Waiter phone (optional: staff use their own)	Samsung Galaxy A06 / A07 / A16	EGP 5,800 - 11,900	A06 5,800-6,899; A07 9,700-11,900; A16 7,499-11,250 (Vodafone, Amazon.eg)	Low: wide spread between sellers
Receipt printer 80 mm (budget)	Xprinter XP-80T USB / XP-N160II USB	EGP 1,899 - 2,750	Amazon.eg 1,899 (XP-80T); Amazon.eg/Noon 2,590-2,750 (XP-N160II)	Medium-high
Receipt printer 80 mm (brand)	Epson TM-T20III USB	EGP 5,999 - 10,499	2B 5,999 (out of stock); Egypt Laptop 6,199-8,699; United-Egypt 7,150; Amazon.eg 7,830-10,000; CairoPOS network 10,499	Medium: big spread, variant matters (-0011 USB, -012 network)
Cash drawer RJ11/RJ12	Alfa ECH405G / GDA RJ-H4 / Zkteco ZKC0508E	EGP 1,299 - 1,899	Kimo Store 1,299-1,600; Trademark Egypt 1,899	Medium: width not stated on listings
UPS 650 VA (printer + router + tablet charger)	APC Easy UPS BV650	EGP 2,449 - 3,850	2B 3,199; Noon 3,250; Egypt Laptop 2,449; MEST 3,850	Medium
Wi-Fi 6 router	TP-Link Archer AX10 / AX23	EGP 2,999 - 4,250	Amazon.eg 3,600; egprices 2,999-4,250; AX23 3,850	Medium
Ceiling access point (per unit)	TP-Link EAP225 (Wi-Fi 5) / EAP610 (Wi-Fi 6)	EGP 3,199 - 4,140	Egypt Laptop 3,199; Amazon.eg 3,555-3,999; EAP610 4,099	Medium
4G backup router	TP-Link TL-MR6400	EGP 2,875 - 3,999	Albadr 2,875; Amazon.eg 3,123; Noon 3,450; Jumia 3,999	Medium
24-inch monitor (kitchen screen)	Samsung / Dell / MSI 24" FHD	EGP 3,375 - 5,799	Noon Dell 3,375; Jumia Samsung 4,850; MSI 5,799	Medium; needs a small Android box or mini-PC: no price found

Not found (no source): printed QR table stands in Egypt (international wholesale is $1.50-1.80 each at 100 pieces; retail $7-16, plus shipping and duties: get a local print quote), an Android TV box price for a kitchen screen, and business-fibre prices from Vodafone or Orange.

9.1 Café starter kits (what a café spends once)

Kit	Contents	Low (EGP)	High (EGP)	USD at EGP 50
Starter (cashier only)	one till: budget tablet, budget printer, drawer. Waiters use their own phones; no kitchen screen (printed tickets or phone)	9,497	11,648	$190 - $233
Recommended (30-table café)	Tab A9+ till, budget printer, drawer, UPS, Lenovo tablet as kitchen screen, 4G backup, two access points	32,518	41,695	$650 - $834
Premium	two tills with Epson printers, kitchen printer, monitor screen, 3 access points, new router	66,739	93,380	$1,335 - $1,868

If you resell the kit: your cost is the retailer price above; a 10-15% handling margin plus delivery and setup is normal. Alternatively sell nothing and publish the kit as a recommended shopping list, which carries no stock risk, no warranty work and no VAT paperwork for you. For a first 10 cafés I recommend the list, not reselling.

9.2 Internet at the café (café pays)

The app is light: a staff tablet plus guests use roughly 0.5 Mbps at peak and about 26 GB a month per busy café on the current polling design (5.1 M requests x 5 KB), far less on option C. What matters is a stable line, not speed. Telecom

Egypt's business-labelled WE Space Super page lists 30 Mbps plans from EGP 210 (140 GB) to EGP 1,360 (1 TB) per month, VAT excluded; consumer WE Space Mega fibre starts at EGP 590 for 70 Mbps and 250 GB. I found no Vodafone or Orange business-fibre price list. **Recommend a fixed line plus a TP-Link MR6400 4G router (EGP 2,875-3,999) with a second operator SIM as backup**; get written quotes for your café addresses.

10. Setup comparison and decision table

Criterion	A: Vercel + Supabase, polling as built	B: same, adaptive polling (recommended now)	C: own Hetzner WebSocket server + Supabase (recommended later)
Monthly cost, 50 busy cafés	$1,702	$474	$53
Monthly cost, 10 busy cafés	$374	$64	$37
Order reaches the kitchen	up to about 2.5 s	about 1-5 s (fast right after an action)	instant (pushed)
Distance from Egypt	Frankfurt (my estimate 40-70 ms, not verified; WonderNetwork has the Cairo-Frankfurt table)	same	Falkenstein / Nuremberg / Helsinki: similar
You operate servers?	No	No	Yes: updates, backups, restarts, monitoring
If it fails	Vercel / Supabase teams fix it	same	You fix it; all cafés on that server are down
Scales with cafés	cost grows linearly	cost grows slowly	cost nearly flat; add a server per ~25-30 busy cafés (my estimate)
Work still to build	none	about one day: slower guest polling, "nothing changed" replies	deployment scripts, process manager, backups, sharding by café; the WebSocket mode already exists in the code
Risk	bill shock as cafés grow	low	medium until you have a runbook
Verdict	Do not scale on it	**Start here**	**Move here at 15-20 busy cafés**

Hosting regions to set: Vercel functions **fra1**; Supabase project in **Frankfurt (eu-central-1)**; if you use Hetzner choose Falkenstein or Nuremberg (the cost-optimised plans were listed for NBG1 and HEL1). Put Cloudflare (free plan) in front of option C for DNS and basic protection. Keep nightly database backups on Supabase Pro (7 days included); add point-in-time recovery ($100 a month) only once cafés depend on it.

11. Final budget to approve (platform + one pilot café)

Item	Who pays	Amount
Platform, first 10 cafés on option B	You	$64 / month
Platform, 50 cafés on option B	You	$474 / month
Platform, 50 cafés on option C	You	$53 / month
Domain	You	about $12 / year (estimate)
AI usage (VIP cafés, Haiku model)	You (include in VIP price)	about $3 / month for 15 cafés
Hardware: recommended kit	Café	EGP 32,518 - 41,695
Internet	Café	EGP 210 - 1,360 / month (WE Space Super, VAT excl.) or a fibre quote
Paymob	Café	2.75% + EGP 3 per card payment
ETA digital seal	Café	EGP 2,000 first year (incl. VAT)
Two test printers (one Xprinter, one Epson) before recommending	You, one-off	EGP 7,898 - 13,249

12. Appendix: source links

Vendor pricing: vercel.com/pricing; vercel.com/docs/pricing/regional-pricing/fra1; supabase.com/pricing; hetzner.com/cloud/cost-optimized; platform.claude.com/docs/en/about-claude/pricing; paymob.com/en/online-payment; orange.eg/en/business/business-solutions/digital-seal-and-digital-signature; cloudflare.com/products/registrar; te.eg (WE Space).

Hardware listings: amazon.eg (Galaxy Tab A9 Plus, Epson TM-T20x, Xprinter XP-N160II, TP-Link Archer AX10, EAP225, EAP610, MR6400); noon.com/egypt-en (Tab A9 Plus, Tab A9, Xprinter, Lenovo Tab M10, UPS, Dell 24" monitor); ennap.com (Tab A9 / A9+); ehabgroup.com (Tab A9); myxprs.com (Lenovo Tab M10); egyptlaptop.com (Epson TM-T20III, EAP225, EAP610, UPS); 2b.com.eg (Epson TM-T20III, APC UPS); united-egypt.net (Epson); cairopos.com (Epson network); kimostore.net (cash drawers); trademark-eg.com (GDA drawer); jumia.com.eg (monitors, MR6400); egprices.com (Archer AX10 / AX23); eshop.vodafone.com.eg (Galaxy A16).

13. Summary, open gaps and next steps

Decision: start on Vercel Pro (fra1) + Supabase Pro (Frankfurt) with adaptive polling; move to your own Hetzner WebSocket server + Supabase at about 15-20 busy cafés. Do not scale on the polling design as built (about $1,702 a month at 50 cafés).

Hardware: recommended café kit EGP 32,518 - 41,695 (starter EGP 9,497 - 11,648, premium EGP 66,739 - 93,380). The café pays the retailer; for the first 10 cafés publish the kit as a shopping list instead of reselling.

Gaps (no source found, get local quotes): printed QR table stands in Egypt; an Android box for a kitchen screen; Vodafone / Orange business-fibre prices. samsung.com/eg showed no tablet prices, so Samsung figures are retailer prices. All retailer prices are listing snapshots: check the live listing before paying.

Assumptions to confirm: EGP 50 per USD (sources disagreed, about 31 versus 51); Hetzner plans showed "currently unavailable", so check stock; load per café is my estimate until a load test is run; Egypt-to-Frankfurt latency (40-70 ms) is not measured.

Not yet tested: the printing code on a real printer. Buy one Xprinter and one Epson and test both with the app before recommending either.

Next steps, in order: (1) build the adaptive-polling change (about one day); (2) run a load test to replace the load estimate; (3) test the two printers; (4) get quotes for QR stands and business internet; (5) confirm the exchange rate with your accountant; (6) pilot with one café, then approve the budget in section 11.